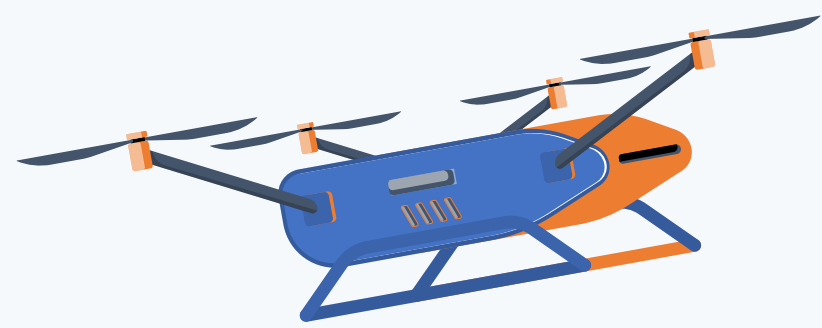


AeroMind: Gesture-Based Drone Control System

Sayed Jihad ElSayed, University of Alabama at Birmingham

Problem

- Traditional drone control requires manual controllers
- Not intuitive for beginners
- High risk of incorrect commands



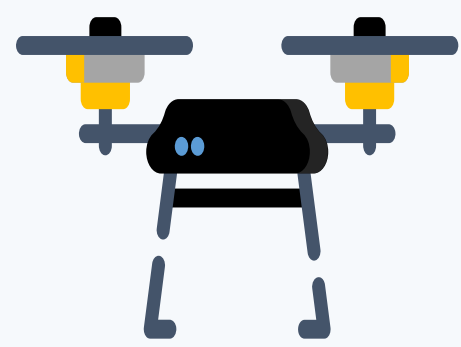
- Complex Control**
Drone controllers require training and are hard to use
- High Learning Curve**
New users struggle to operate drones safely
- Unstable Gestures**
Gesture recognition is noisy and inconsistent
- Safety Risk**
Wrong commands can lead to crashes

Solution

- Real-time gesture-based drone control using camera input
- Uses hand tracking and classification
- Includes safety filters and command cooldown system



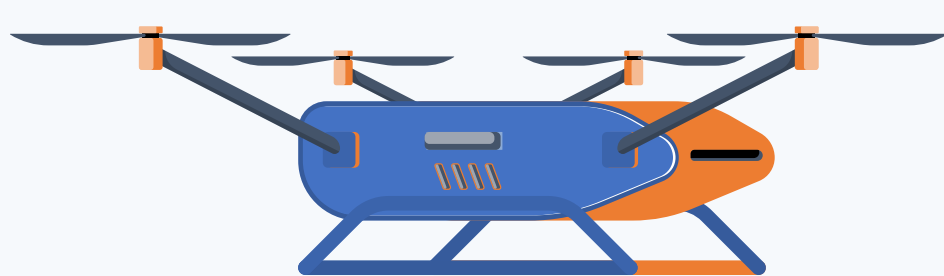
Gesture Control
Hand gestures replace traditional controllers



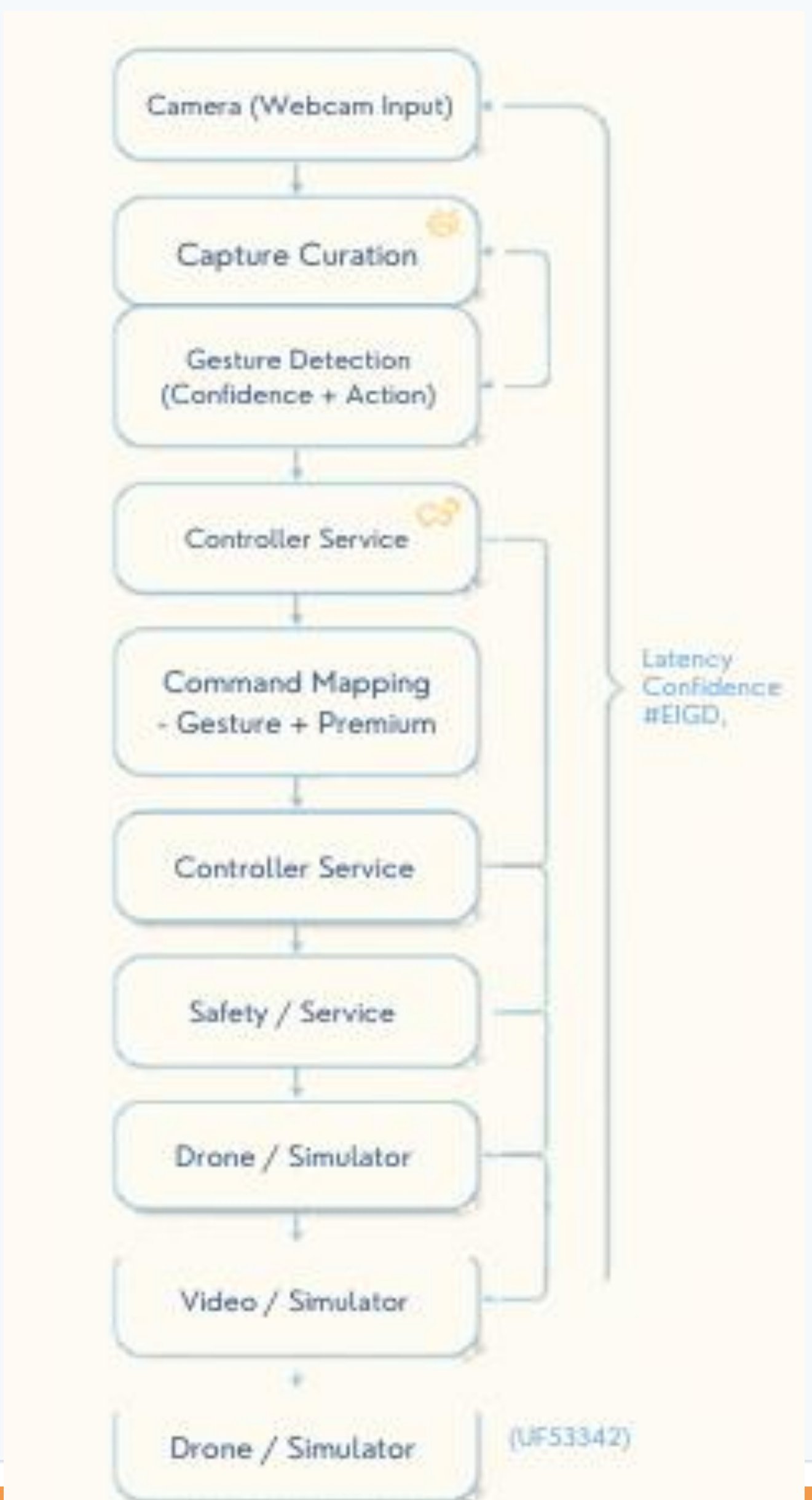
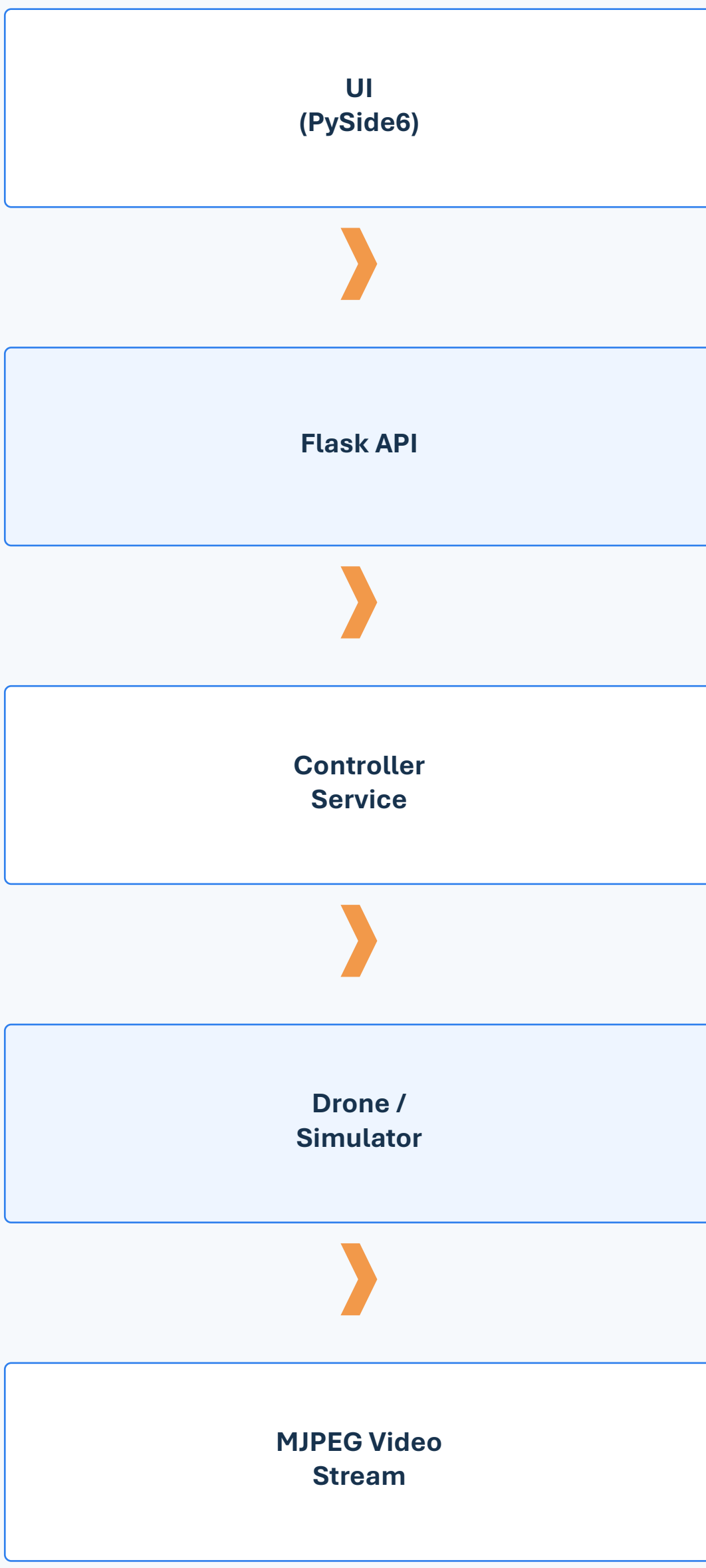
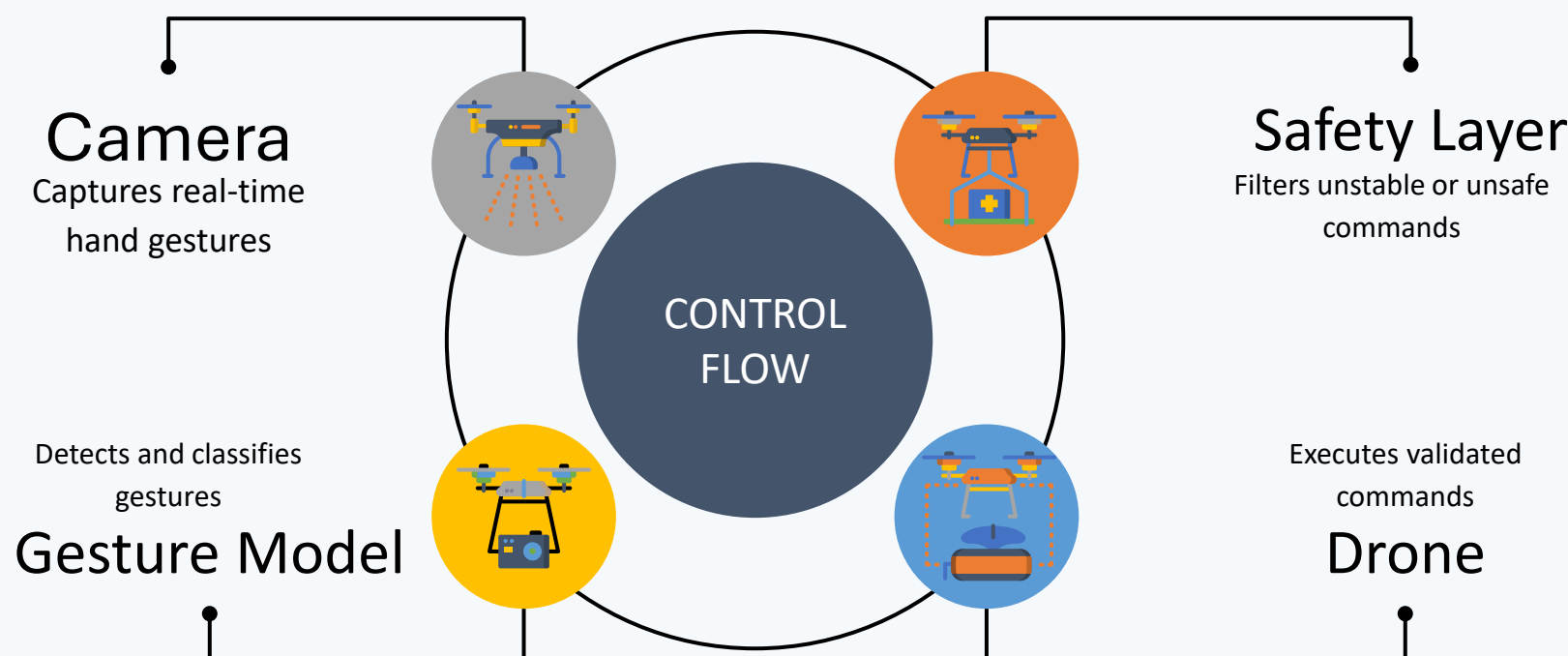
AI + Safety Layer
Filters unstable commands before execution

Key Features

- Hand gesture recognition (MediaPipe)
- One-shot and repeatable commands
- Safety blocking system
- Real-time video feedback



System Architecture



Results

- Command dispatch rate and blocked rate
- Improved stability with cooldown logic
- Reduced false commands

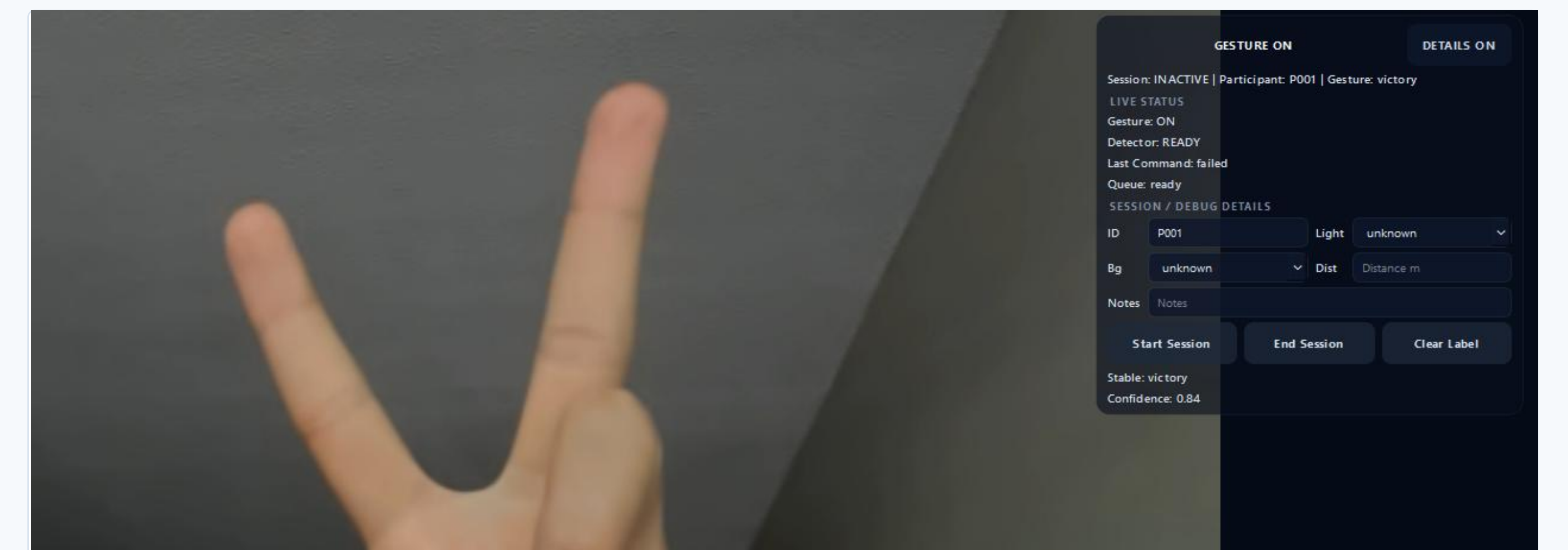
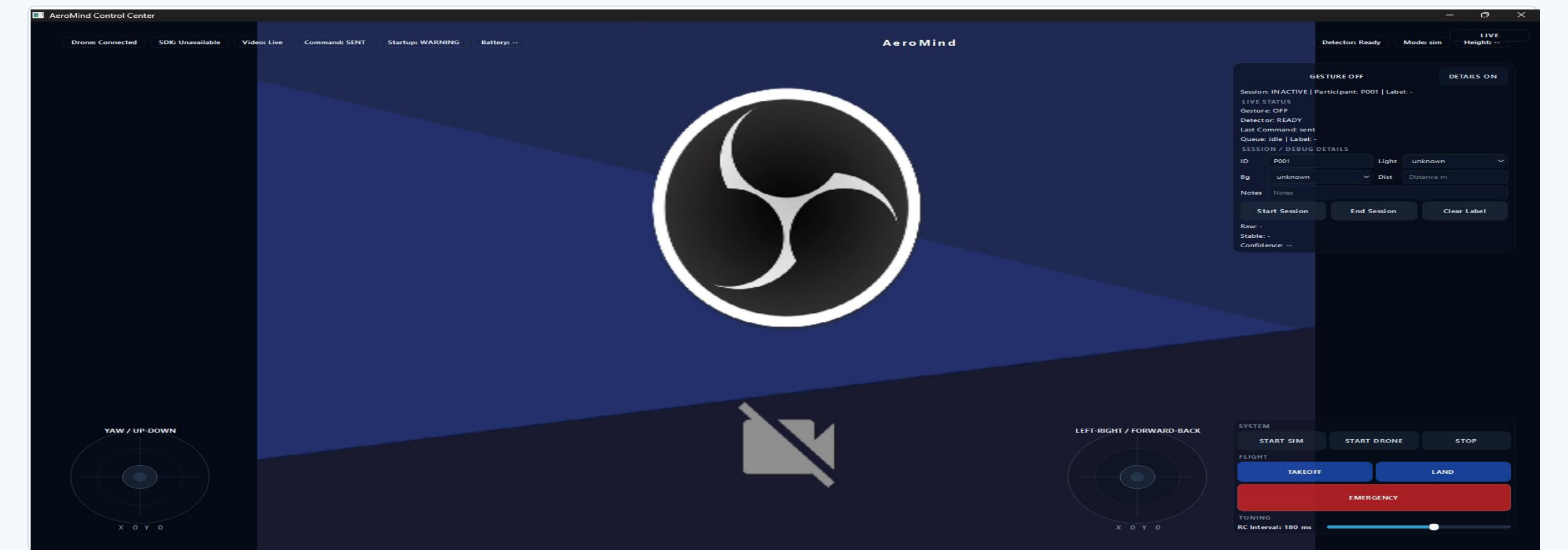
Latency: 0.40s
Avg Confidence: 0.82
Commands Blocked: 60%
Reason: Safety filter

Metrics to highlight in final print

- Blocked-command percentage
- Command latency trend
- Stability gain after cooldown tuning

0.60 Blocked Commands
Safety layer filters unstable inputs

Demo / Visuals



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y		
1	Run ID	ts	event_type	frame_id	participant	lighting	background	distance	/gesture_trig	gesture_pr	stable_ges	confidence	state_ms	stable	hit1	frame_c1	inference1	t_stable	rt1	command1	t_command	api_round	vision_to_t	t_stable_to_t	total_client	threshold	resolved_c
2	2.02044E-17	7801E+00	command	63	P001	unknown	unknown	-	-	-	-	0.66	1	1.19E+09	1.19E+09	unknown	-	-	-	-	-	-	-	34	0.6	-	
3	2.02044E-17	7801E+00	command	64	P001	unknown	unknown	-	-	-	-	0.94	0	1.19E+09	1.19E+09	unknown	-	-	-	-	-	-	-	39	0.6	-	
4	2.02044E-17	7801E+00	command	127	P001	unknown	unknown	-	-	-	-	0.94	1	1.19E+09	1.19E+09	unknown	-	-	-	-	-	-	-	34	0.6	-	
5	2.02044E-17	7801E+00	gesture_re	128	P001	unknown	unknown	-	-	-	-	0.94	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	19	43	1	63	0.72	hover	38	0.72	hover	
6	2.02044E-17	7801E+00	command	128	P001	unknown	unknown	-	-	-	-	0.94	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	19	43	1	63	0.72	hover	38	0.72	hover	
7	2.02044E-17	7801E+00	command	129	P001	unknown	unknown	-	-	-	-	0.95	0	1.19E+09	1.19E+09	1.19E+09	1.19E+09	39	39	0.72	hover	38	0.72	hover			
8	2.02044E-17	7801E+00	command	130	P001	unknown	unknown	-	-	-	-	0.95	0	1.19E+09	1.19E+09	1.19E+09	1.19E+09	25	25	0.6	-	-	-	-	-		
9	2.02044E-17	7801E+00	gesture_re	148	P001	unknown	unknown	-	-	-	-	0.94	1	1.19E+09	1.19E+09	unknown	-	-	-	-	-	-	-	51	0.6	-	
10	2.02044E-17	7801E+00	command	149	P001	unknown	unknown	-	-	-	-	0.94	0	1.19E+09	1.19E+09	unknown	-	-	-	-	-	-	-	28	0.6	-	
11	2.02044E-17	7801E+00	command	198	P001	unknown	unknown	-	-	-	-	0.94	1	1.19E+09	1.19E+09	unknown	-	-	-	-	-	-	-	32	0.6	-	
12	2.02044E-17	7801E+00	gesture_re	199	P001	unknown	unknown	-	-	-	-	0.94	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	38	38	0.72	hover	38	0.72	hover			
13	2.02044E-17	7801E+00	command	199	P001	unknown	unknown	-	-	-	-	0.94	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	38	38	0.72	hover	38	0.72	hover			
14	2.02044E-17	7801E+00	command	201	P001	unknown	unknown	-	-	-	-	0.94	0	1.19E+09	1.19E+09	unknown	-	-	-	-	-	-	-	37	0.6	-	
15	2.02044E-17	7801E+00	command	202	P001	unknown	unknown	-	-	-	-	0.94	0	1.19E+09	1.19E+09	unknown	-	-	-	-	-	-	-	23	0.6	-	
16	2.02044E-17	7801E+00	command	209	P001	unknown	unknown	-	-	-	-	0.95	1	1.19E+09	1.19E+09	unknown	-	-	-	-	-	-	-	37	0.6	-	
17	2.02044E-17	7801E+00	gesture_re	210	P001	unknown	unknown	-	-	-	-	0.95	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	34	34	0.78	takeoff	34	0.78	takeoff			
18	2.02044E-17	7801E+00	command	210	P001	unknown	unknown	-	-	-	-	0.95	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	34	34	0.78	takeoff	34	0.78	takeoff			
19	2.02044E-17	7801E+00	command	213	P001	unknown	unknown	-	-	-	-	0.95	284	3	1.19E+09	1.19E+09	1.19E+09	1.19E+09	19	44	1	64	0.78	takeoff			
20	2.02044E-17	7801E+00	command	214	P001	unknown	unknown	-	-	-	-	0.95	355	3	1.19E+09	1.19E+09	1.19E+09	1.19E+09	37	37	0.78	takeoff	37	0.78	takeoff		
21	2.02044E-17	7801E+00	gesture_re	213	P001	unknown	unknown	-	-	-	-	0.95	355	3	1.19E+09	1.19E+09	1.19E+09	1.19E+09	19	44	1	64	0.78	takeoff			
22	2.02044E-17	7801E+00	command	242	P001	unknown	unknown	-	-	-	-	0.94	2809	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	37	37	0.78	takeoff	37	0.78	takeoff		
23	2.02044E-17	7801E+00	command	243	P001	unknown	unknown	-	-	-	-	0.88	2892	1	1.19E+09	1.19E+09	unknown	-	-	-	-	-	33	0.6	-		
24	2.02044E-17	7801E+00	gesture_re	244	P001	unknown	unknown	-	-	-	-	0.88	2894	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	40	40	0.72	hover	40	0.72	hover		
25	2.02044E-17	7801E+00	command	244	P001	unknown	unknown	-	-	-	-	0.88	2894	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	26	40	1	67	0.72	hover			
26	2.02044E-17	7801E+00	command	245	P001	unknown	unknown	-	-	-	-	0.94	3167	0	1.19E+09	1.19E+09	unknown	-	-	-	-	-	35	0.6	-		
27	2.02044E-17	7801E+00	command	246	P001	unknown	unknown	-	-	-	-	0.94	3162	1	1.19E+09	1.19E+09	unknown	-	-	-	-	-	35	0.6	-		
28	2.02044E-17	7801E+00	gesture_re	247	P001	unknown	unknown	-	-	-	-	0.94	3240	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	37	37	0.72	hover	37	0.72	hover		
29	2.02044E-17	7801E+00	command	247	P001	unknown	unknown	-	-	-	-	0.94	3240	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	17	37	2	56	0.72	hover			
30	2.02044E-17	7801E+00	command	248	P001	unknown	unknown	-	-	-	-	0.88	3323	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	35	35	0.72	hover	35	0.72	hover		
31	2.02044E-17	7801E+00	gesture_re	249	P001	unknown	unknown	-	-	-	-	0.88	3403	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	29	29	0.72	hover	29	0.72	hover		
32	2.02044E-17	7801E+00	command	249	P001	unknown	unknown	-	-	-	-	0.88	3403	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	4	29	2	35	0.72	hover			
33	2.02044E-17	7801E+00	command	250	P001	unknown	unknown	-	-	-	-	0.88	3495	3	1.19E+09	1.19E+09	1.19E+09	1.19E+09	35	35	0.72	hover	35	0.72	hover		
34	2.02044E-17	7801E+00	gesture_re	252	P001	unknown	unknown	-	-	-	-	0.94	354	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	36	36	0.72	hover	36	0.72	hover		
35	2.02044E-17	7801E+00	gesture_re	253	P001	unknown	unknown	-	-	-	-	0.94	348	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	35	35	0.72	hover	35	0.72	hover		
36	2.02044E-17	7801E+00	command	253	P001	unknown	unknown	-	-	-	-	0.94	348	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	19	35	1	55	0.72	hover			
37	2.02044E-17	7801E+00	command	254	P001	unknown	unknown	-	-	-	-	0.94	434	3	1.19E+09	1.19E+09	1.19E+09	1.19E+09	34	34	0.72	hover	34	0.72	hover		
38	2.02044E-17	7801E+00	command	259	P001	unknown	unknown	-	-	-	-	0.88	531	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	42	42	0.72	hover	42	0.72	hover		
39	2.02044E-17	7801E+00	command	260	P001	unknown	unknown	-	-	-	-	0.94	611	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	36	36	0.72	hover	36	0.72	hover		
40	2.02044E-17	7801E+00	command	262	P001	unknown	unknown	-	-	-	-	0.88	804	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	55	55	0.72	hover	55	0.72	hover		
41	2.02044E-17	7801E+00	command	263	P001	unknown	unknown	-	-	-	-	0.94	853	2	1.19E+09	1.19E+09	1.19E+09	1.19E+09	44	44	0.72	hover	44	0.72	hover		
42	2.02044E-17	7801E+00	command	268	P001	unknown	unknown	-	-	-	-	0.66	1301	3	1.19E+09	1.19E+09	1.19E+09	1.19E+09	34	34	0.72	-	-	-	-		
43	2.02044E-17	7801E+00	gesture_re	269	P001	unknown	unknown	-	-	-	-	0.94	1391	3	1.19E+09	1.19E+09	1.19E+09	1.19E+09	38	38	0.72	hover	38	0.72	hover		
44	2.02044E-17	7801E+00	command	269	P001	unknown	unknown	-	-	-	-	0.94	1391	3	1.19E+09	1.19E+09	1.19E+09	1.19E+09	38	38	0.72	hover	38	0.72	hover		

Conclusion

- Gesture control is feasible for drone systems
- Future improvements needed in accuracy and latency



LIVE DEMO